

Here's the compressed version of the notes, condensed to fit within 2 printed pages (max 500 words):

Python Basics

* Definition: Python is a high-level, multi-purpose programming language known for its simplicity and versatility.

* Key Points:

- + Fastest-growing and most popular programming language
- + Ideal for beginners due to simple syntax and ease of use
- + Wide range of applications, including data analysis, AI, machine learning, web development, automation, etc.
- + Huge community and large ecosystem of libraries, frameworks, and tools

* Short Answer: Python is a versatile language that can be used in various fields. Its simplicity and ease of use make it an ideal choice for beginners.

Possible Exam Questions:

1. What are the key features of the Python programming language?
2. Why is Python considered an ideal language for beginners?
3. List some areas where Python can be applied in real-world scenarios.

Variables, Strings, and String Methods

* Definition: Variables store data in computer's memory.

* Key Points:

- + Variables can hold different types of data (numbers, booleans, strings)
- + Strings can be enclosed in single quotes or double quotes
- + Escape sequences represent special characters in strings
- + Formatted strings build strings using expressions

* Short Answer: A variable is a label that refers to a memory location where data is stored.

Possible Exam Questions:

1. What is the difference between a string enclosed in single quotes and double quotes?

2. How do you represent a newline character in a string using escape sequences?
3. What is the purpose of the ``len()`` function in Python?

String Methods

* Definition: String methods perform various operations on strings.

* Key Points:

- + ``upper()``: Returns string in uppercase
- + ``lower()``: Returns string in lowercase
- + ``title()``: Capitalizes first letter of every word
- + ``strip()``, ``lstrip()``, ``rstrip()``: Remove white space from strings
- + ``find()``: Returns index of character or sequence of characters
- + ``replace()``: Replaces character or sequence of characters with

another string

* Short Answer: A string method is used to manipulate and analyze strings.

Possible Exam Questions:

1. What is the output of ``print("Hello World".upper())``?
2. How do you remove white space from both ends of a string using Python?
3. What is the difference between the ``find()`` and ``in Operator`` methods?

Functions

* Definition: A function performs a specific task, which can be executed multiple times.

* Key Points:

- + Functions return values by default (None) unless specified otherwise
- + Functions can carry out tasks or calculate and return values
- + All functions have parameters, which are required by default but can

be made optional

* Short Answer: A function is defined with its own block of code and can return values or carry out tasks.

Possible Exam Questions:

1. What is returned by a function if no value is specified?
2. How do you make a parameter optional in a function?
3. What is the purpose of using an asterisk (*) before a parameter name?